

ISO 13485:2016

Medical Devices - Quality management systems - Requirements for regulatory purposes

Assessment 1

- Lesson 1: Why do we need a Quality Management System?
- Lesson 2: Scope and Applicability of ISO 13485:2016

Question 1:

True or false?

A Quality Management System is a set of tools and methods used to assure the quality, effectiveness and safety of a company's products or services.

Solution: True

Question 2:

True or false?

Using a QMS can make the organization be more competitive, reduce its running costs, increase revenue and improve customer's satisfaction.

Solution: True

Question 3:

True or false?

Meeting the expected customer or regulatory requirements is not an objective of ISO 13485.

Solution: False

Question 4:

Select all correct answers.

What are the characteristics of a Quality Management System compliant to ISO 13485?

(True) It enables different stakeholders within the industry to interact more efficiently.

(False) It reduces the need of documenting the critical processes as QMS policies and procedures.

(True) It allows companies in the medical devices industry to sell their products and services worldwide.

(False) The organization must implement and meet all requirements of ISO 13485 to comply with the standard.

(True) It requires to establish a risk-based approach to the applicable processes.

(True) It requires to implement, control, maintain and improve all applicable documents, operations, processes and resources for continuous regulatory compliance.

Assessment 2

- Lesson 3: General QMS requirements.
- Lesson 4: Documentation requirements.

Question 1:

True or false?

Any relevant requirement, process, activity, arrangement, or procedure shall be identified, included, planned, implemented, and controlled in the QMS.

Solution: True

Question 2:

True or false?

The activities and controls the organization must implement in its QMS are independent of the role of the organization (manufacturer, authorized representative, importer, or distributor).

Solution: False

Question 3:

True or false?

The organization must determine the criteria and methods for ensuring that both the operation and control of processes of the QMS are effective.

Solution: True

Question 4:

True or false?

Records of the QMS processes can be created to demonstrate compliance to ISO 13485 and other applicable regulatory requirements, but they don't need to be stored or maintained.

Solution: False

Question 5:

True or false?

If the organization decides to outsource some processes to an external vendor, it is the external vendor's responsibility to ensure that those processes are monitored and controlled, if they can impact the conformity of the products or services.

Solution: False

Question 6:

True or false?

Quality Agreements can be verbally negotiated and agreed between the organization and the external vendors.

Solution: False

ISO 13485:2016

Medical Devices - Quality management systems - Requirements for regulatory purposes

Question 7:

True or false?

Computer software applications used to support the QMS activities must be validated prior to initial use, following a validation procedure.

Solution: True

Question 8:

True or false?

Changes to any process or computer software shall be controlled and evaluated based on risk.

Solution: True

Question 9:

Select all correct answers.

What should the Quality Manual include?

(True) The scope of the quality management system, including a list of the exclusions and nonapplications, and their justification.

(False) The product user manuals and other instructions for use.

(True) The list of relevant QMS procedures, or a reference to them.

(True) A description of the interactions between the processes of the quality management system.

(False) Any relevant financial information related to the product of services provided by the organization.

(True) A description of the structure of the documentation system used by the quality management system.

Question 10:

True or false?

The organization must develop a Medical Device File, specific to each product model, type, class, or category. The file must include the relevant documents describing the medical device specifications, and realization, servicing and installation processes.

Solution: True

Question 11:

True or false?

New or updated documentation and records necessary for the operation of the quality management system should be released for use as soon as possible, without review, to reduce risk.

Solution: False

Question 12:

True or false?

Records shall be retained for at least the lifetime of their relevant medical device, according to the manufacturer's specifications, but not less than 2 years since the medical device was released to the market (or as described by other applicable regulatory requirements).

Solution: False

Assessment 3

- Lesson 5: Management responsibility
- Lesson 6: Resource management

Question 1:

Select all correct answers.

Per ISO 13485, the top management team must provide evidence of its commitment to the development, implementation and effectiveness of the QMS, as follows:

(True) Communicating to the organization the importance of meeting customers' and regulatory requirements.

(True) Documenting the Quality Policy and Quality Objectives.

(False) Ensuring the financial results are meeting the revenue targets.

(True) Conducting Management Reviews, to verify the effectiveness of the QMS.

(False) Allocating funds to the marketing and sales team.

(True) Ensuring the availability of resources.

Question 2:

True or false?

Once defined and established, the Quality Policy doesn't need to be periodically reviewed or updated.

Solution: False

Question 3:

True or false?

Top management must ensure that the integrity of the quality management system is not impacted when changes are planned and implemented.

Solution: True

Question 4:

True or false?

Top management shall appoint a management representative, with the authority and responsibility to report on the effectiveness of the QMS, and on opportunities for improvements.

Solution: True

ISO 13485:2016

Medical Devices – Quality management systems – Requirements for regulatory purposes

Question 5:

Select all correct answers.

The purpose of the Management Review process is to:

(True) Review the QMS to ensure its continuous suitability and effectiveness.

(True) Assess opportunities for improvement of the QMS.

(False) Assess the financial results of the company.

(True) Assess the need for changes to the Quality Policy and Quality Objectives.

Question 6:

True or false?

The inputs and outputs of the Management Review process don't need to be recorded or maintained.

Solution: False

Question 7:

True or false?

Personnel performing any activity that may affect product quality must be competent in terms of education, training, skills, and experience.

Solution: True

Question 8:

True or false?

Control of personnel health, cleanliness and clothing is not a requirement of ISO 13485.

Solution: False

Question 9:

True or false?

The organization shall document the requirements for the infrastructure required to achieve conformity to product requirements and to prevent product mix-up and ensure orderly handling of product.

Solution: True

Question 10:

True or false?

ISO 13485 refers to standards ISO 14644 and ISO 14698 for additional information regarding Cleanrooms and associated controlled environments.

Solution: True

Assessment 4

▪ Lesson 7: Product realization (Section 1)

Question 1:

True or false?

A Process Failure Mode and Effects Analysis (pFMEA) is a risk management method that can be used to identify and mitigate risks related to a specific process or set of activities.

Solution: True

Question 2:

True or false?

As part of product realization planning, the organization shall establish the required verification, validation, monitoring, measurement, inspection and test, material handling and storage, distribution and traceability activities to be used, and the criteria to be applied for the acceptability of product.

Solution: True

Question 3:

True or false?

Evidence of the realization of product meeting all applicable customer and regulatory requirements can be captured in production records, but that requirement is optional.

Solution: False

Question 4:

Select all correct answers.

Per ISO 13485, the organization shall determine:

(False) The expected market share and revenue of the products or services to be provided.

(True) The product requirements as defined by the customer, including delivery and post-delivery activities.

(True) Additional product requirements that may not explicitly defined by the customer, but are needed based on the intended use of the product.

(True) The regulatory requirements applicable to the product.

(False) The taxes to be paid by the product importers or distributors.

(True) The user training required to ensure the safe and effective use of the product.

Question 5:

True or false?

User training doesn't need to be provided by the organization if it is determined that lack of training would not affect the safe and effective use of the product.

Solution: True

ISO 13485:2016

Medical Devices - Quality management systems - Requirements for regulatory purposes

Question 6:

True or false?

All product requirements identified must be reviewed by the organization, prior to committing to supply those products to the customer.

Solution: True

Question 7:

True or false?

Once the product requirements are defined and documented, they cannot be modified within the lifetime of the product.

Solution: False

Question 8:

True or false?

Communication with customers shall be planned and arranged following a documented process, when sharing information like product information, enquiries, contracts and product ordering, customer feedback and complaints handling or advisory notices.

Solution: True

Question 9:

Select all correct answers.

The input requirements to the Design and Development process shall be:

(False) General and open to interpretation.

(True) Complete and unambiguous.

(True) Able to be tested (i.e., verified or validated).

(True) Not in conflict with each other.

Question 10:

True or false?

Design Verification shall be executed to confirm the design outputs meet the input requirements.

Solution: True

Question 11:

True or false?

Design Validation shall be executed to confirm the resulting product is capable of meeting the specified intended use.

Solution: True

Question 12:

True or false?

Design Validation can be executed by the engineering department of the organization, without feedback from customers or clinicians.

Solution: False

Question 13:

True or false?

Transfer activities of design and development outputs to manufacturing shall ensure that those outputs are suitable for manufacturing, before becoming final production specifications, and that production capability can meet product requirements.

ISO 13485:2016

Medical Devices - Quality management systems - Requirements for regulatory purposes

Solution: True

Question 14:

True or false?

Design and development changes shall be identified, reviewed, verified, validated (as applicable) and approved.

Solution: True

Assessment 5

▪ Lesson 7: Product realization (Section 2)

Question 1:

True or false?

The organization shall maintain documented procedures to ensure that purchased product (material, components, etc.) conforms to specifications

Solution: True

Question 2:

True or false?

The organization shall assess and approve potential suppliers. Once approved, the suppliers don't need to be monitored or re-evaluated by the organization.

Solution: False

Question 3:

True or false?

Purchasing information shall describe the purchased product, including specifications and requirements for product acceptance, processes, procedures, and the use of tools and equipment.

Solution: True

Question 4:

True or false?

The suppliers shall notify the organization of changes in the purchased product, if those changes can affect the ability of the purchased product to meet specified purchase requirements

Solution: True

Question 5:

True or false?

To ensure product conformity to specifications, the production and service provision shall be planned, carried out, monitored and controlled. Production controls don't need to include the qualification of infrastructure.

Solution: False

Question 6:

True or false?

If servicing of the medical device is a specified requirement, the organization shall document servicing procedures, materials and measurements, required to complete those activities and verify that product requirements are met.

Solution: True

Question 7:

Select all correct answers.

Procedures to perform process validations shall be documented, including:

(True) Defined criteria for review and approval of the processes.

(True) Qualification of equipment and personnel.

(True) Use of methods, procedures, and acceptance criteria.

(False) Customer complaints.

(True) Statistical techniques with rationale for sample sizes, as applicable.

(True) Requirements for validation records.

(True) Revalidation and criteria for revalidation.

(True) Approval of changes to the processes.

Question 8:

True or false?

The organization shall document procedures for the validation of processes for sterilization and for sterile barrier systems, and for their revalidation following product or process changes, as appropriate.

Solution: True

Question 9:

Select the correct answer.

The status of products, regarding their conformity to monitoring and measurement requirements, shall be identified and maintained:

(True) Throughout product realization, including production, storage, installation, and service and repair activities.

(False) Only at the end of the manufacturing process, prior to releasing the product for distribution.

Question 10:

True or false?

To ensure valid results, the monitoring and measuring equipment shall be calibrated or verified prior to its initial use, at specified intervals, against measurement standards traceable to international or national standards.

Solution: True

Question 11:

Select the correct answer.

If monitoring and measuring equipment is found not to conform to requirements, the organization shall:

(False) Replace the faulty equipment and continue testing.

(True) Assess and record the validity of the previous measuring results, and take appropriate actions regarding the equipment and products affected.

Assessment 6

▪ Lesson 8: Measurement, analysis and improvement

Question 1:

True or false?

The organization shall plan and implement processes for monitoring, measurement, analysis and improvement, required to maintain the effectiveness of the quality management system.

Solution: True

Question 2:

Select the correct answer.

A feedback process shall be established, based on data from:

(False) Post-production activities only.

(True) Production and post-production activities.

Question 3:

True or false?

The complaints handling process shall define the requirements and responsibilities for receiving and recording, evaluating and investigating complaints.

Solution: True

Question 4:

True or false?

Internal audits shall be conducted by the organization at planned intervals, to confirm the QMS conforms to the requirements of the ISO 13485 standard and other applicable regulatory requirements, and the QMS is effectively implemented and maintained.

Solution: True

Question 5:

True or false?

Auditors can audit their own work.

Solution: False

Question 6:

True or false?

Product release and service delivery shall not proceed until all the planned and documented activities have been successfully completed

Solution: True

Question 7:

Select all correct answers.

When nonconforming product is detected prior to delivery, the organization shall take the required actions to:

(True) Eliminate the detected nonconformity.

(True) Prevent the original intended use or application of the product.

(False) Adjust the acceptance criteria to ensure the product passes the tests.

(True) If applicable, authorize the use, release, or acceptance of the product under concession, if justification is provided and approved, and applicable regulatory requirements are met.

Question 8:

True or false?

When nonconforming product is detected after delivery or use has started, the organization shall take the required actions, based on the effects or potential effects of the nonconformity, like issuing advisory notices.

Solution: True

Question 9:

True or false?

The organization shall perform rework of nonconforming product following documented procedures, considering the potential adverse effects of the rework on the product. Rework procedures don't need to be reviewed or approved.

Solution: False

Question 10:

True or false?

Data analysis shall be used as input to improvement activities, when it shows that the quality management system is not suitable, adequate, or effective. The analysis shall include data gathered from various monitoring and measuring activities, such as feedback, conformity to product requirements, characteristics and trends of processes and products, opportunities for improvement, suppliers data, audits data and service reports.

Solution: True

Question 11:

True or false?

Corrective and preventive actions (CAPA process) shall be used to eliminate the causes of potential nonconformities and prevent they occurrence, and also to eliminate the cause of any nonconformity and prevent its reoccurrence.

Solution: True